Introduction to Causal Inference

Final Report:
Does Sustained Daily Cigarette Smoking Throughout Pregnancy
Causally Increase the Risk of Infant Mortality?

July 16, 2026

1 Causal Question

Does sustained daily cigarette smoking throughout pregnancy causally increase the probability of infant death before age one?

Using the potential outcomes framework, let $Y^{(1)}$ denote the infant mortality outcome if a mother smoked throughout her pregnancy, and $Y^{(0)}$ the outcome if she never smoked. The primary estimand is the Average Treatment Effect (ATE): $\tau = E[Y^{(1)} - Y^{(0)}]$. We also report the ATT from propensity score matching, which focuses on mothers who actually smoked rather than the full population.

Treatment. $T = 1$ if the mother reported at least one cigarette per day, on average, in both the pre-pregnancy period and every trimester (`CIG0_R`–`CIG3_R` each in categories 1–5). $T = 0$ requires zero cigarettes in every period and `CIG_REC = N`. Partial quitters and records with unknown tobacco status (`F_TOBACCO` $\neq$ `1`) or any unknown recode (code 6) are dropped. This leaves two clean, well-separated groups.

Outcome. $Y = 1$ if the infant died before their first birthday, from the death-linkage indicator `FLGND = 1`; otherwise $Y = 0$.

Temporal ordering. Smoking is recorded at delivery, before the infant is born. Reverse causality is impossible here by design.

2 Existing Knowledge

The link between prenatal smoking and infant death has been studied for decades. Dietz et al. [1] put numbers to it: roughly 5–7% of US infant deaths are attributable to maternal smoking, about one in fourteen. The mechanism is well understood. Nicotine constricts the blood vessels feeding the placenta, carbon monoxide competes with oxygen in fetal haemoglobin, and the net result is hypoxia, restricted growth, and a sharply elevated risk of preterm birth and abruption [2]. Wisborg et al. [3] documented a dose-dependent relationship with SIDS specifically (adjusted OR ≈ 3.5), and Salihu and Wilson [4] showed the effect varies by dose and cause of death.

A 2023 nationwide cohort by Yuan et al. [7] found ORs ranging from 1.24 among light smokers to 1.81 among heavy smokers after adjusting for sociodemographic factors. The problem is that Yuan et al. also adjust for birth weight and gestational age. Those are not confounders; smoking *causes* low birth weight and prematurity, which then raise mortality. Conditioning on them blocks part of the causal pathway and pulls the estimate toward zero. It is a common mistake in perinatal epidemiology.

Beyond that specific issue, none of the existing work frames this as a causal inference problem in any formal sense. There is no explicit estimand, no overlap check, no doubly-robust estimator, and no sensitivity analysis for unmeasured confounding. The studies establish a strong association with good biological backing, but they cannot tell us how much of that association is causal.

3 Data

We use the **2015 Cohort Linked Birth/Infant Death Data Set** [6] from the National Center for Health Statistics (NCHS). The file covers roughly 3.99 million US births from 2015, tracking each infant through their first birthday. Of those, 23,357 are linked to a death certificate (99.4% linkage rate). The source is administrative: most variables, including the death linkage, age, education, insurance, and clinical conditions, come from official records submitted to NCHS under the Vital Statistics Cooperative Program. The one exception is smoking. Those items are self-reported on the birth certificate at delivery, which matters for interpretation, and Section 7 runs a misclassification sensitivity check specifically because of it.

Analytic sample. Applying all filters (complete tobacco reporting, the treatment/control definition above, non-missing confounders, valid BMI recode 1–6, and known parity) leaves **3,005,458 births**: 147,963 sustained smokers (4.92%) and 2,857,495 never-smokers. Sample size is not the issue here. With three million records the CIs are driven by propensity weight behaviour, not statistical power.

Observed confounders:

Variable	Field	Role
Mother's education	`MEDUC`	SES proxy; predicts both smoking and mortality
Mother's age	`MAGER`	Younger mothers smoke more; age affects risk
Race/Hispanic origin	`MRACEHISP`	Differential smoking prevalence; healthcare access
Marital status	`DMAR`	Social support and economic resource proxy
WIC participation	`WIC`	Low-income nutrition programme; SES proxy
Payment/insurance	`PAY_REC`	Medicaid vs. private; SES and access indicator
Pre-pregnancy diabetes	`RF_PDIAB`	Pre-existing health risk factor
Pre-pregnancy hypert.	`RF_PHYPE`	Pre-existing health risk factor
Pre-pregnancy BMI	`BMI_R`	6-category recode (underweight to extreme obesity)
Father's education	`FEDUC`	Household SES proxy
Father's age group	`FAGE11`	Household SES proxy
Parity	`PRIORLIVE`	Prior live births (0–5+); risk and SES proxy

Mediators excluded. Birth weight (`BRTHWGT`) and gestational age (`COMBGEST`) are left out of all adjustment sets. Smoking causes low birth weight and prematurity, which then raise mortality risk, so including them would partially block the pathway we are trying to measure. We estimate the *total* effect.

Unobserved confounders. Alcohol use, illicit drugs, maternal mental health, and neighbourhood SES are not in the birth certificate data. Section 7 quantifies how much confounding of that type it would take to explain away our results.

4 Causal Graph

Figure 1 shows the proposed DAG. Each observed confounder has edges pointing to both treatment T and outcome Y, opening backdoor paths that adjustment must close. Unobserved confounders (dashed) do the same, which is why we cannot just condition our way out of the problem and need sensitivity analyses instead. Birth weight and gestational age sit on the causal path between T and Y: smoking causes them, and they cause mortality. Including them in a regression would partially block the effect we are trying to estimate, so they are left out.

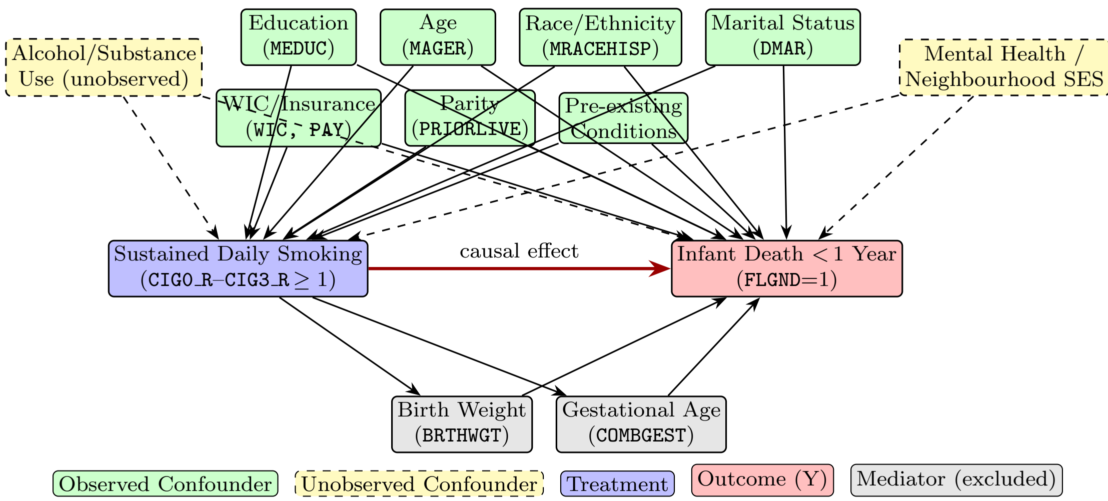

Figure 1: Proposed DAG. Observed confounders (green) form the adjustment set. Alcohol/substance use and mental health (yellow, dashed) are the primary unmeasured confounders. Birth weight and gestational age (grey) are post-treatment mediators excluded from adjustment to estimate the total causal effect. Each confounder has edges to both treatment and outcome, encoding the backdoor paths closed by conditioning on $\mathbf{X}$.

5 Methods and Causal Assumptions

5.1 Causal Assumptions

Consistency. $Y^{(t)} = Y$ whenever $T = t$. Our treatment is binary, where partial quitters are excluded. The main biological mechanisms (nicotine vasoconstriction, fetal hypoxia) do not depend on cigarette brand or delivery method, therefore, the version-of-treatment problem is limited. We would like to note that mothers do smoke different amounts. Our binary definition averages over that variation, and the dose-response sensitivity check in Section 7 shows the effect increases monotonically with dose, which is what we would expect if the causal story is right.

SUTVA. One mother's smoking behaviour cannot plausibly affect another infant's mortality. We could argue that second-hand smoke exposure from other parents might constitute a form of interference, but we consider this effect to be very unlikely or at least negligible in this context.

Conditional ignorability. We assume $T \perp\!\!\!\perp (Y^{(0)}, Y^{(1)}) \mid \mathbf{X}$. Unverifiable, but the dataset covers most of what we would want: both parents' education, insurance type, WIC participation, maternal health history, parity, age, race, and marital status. The main gaps are alcohol use and mental health, which is what the sensitivity analyses are for.

Positivity. Propensity scores range up to 0.85 on both sides, so nominal positivity holds. The problem is the large cluster of high-education, high-income never-smokers with scores near zero. For those women, sustained smoking is practically impossible, and they get enormous IPW weights (max 492; treated ESS 1.4%). The trimming analyses in Section 7 deal with this.

5.2 Estimation Methods

We run five estimators. Four target the ATE; PS matching targets the ATT.

S-Learner. Single logistic regression of Y on T and $\mathbf{X}$. Potential outcomes predicted at $T = 1$ and $T = 0$ for each unit, difference averaged. Simple, stable, but can shrink the treatment coefficient.

T-Learner. Separate logistic models for each arm. Allows the covariate–outcome relationship to differ between smokers and non-smokers.

IPW (Stabilised Hajek). Units reweighted by stabilised inverse-probability weights to create a balanced pseudo-population. Sensitive to extreme weights; its performance here is poor, as the diagnostics show.

AIPW / Doubly Robust (primary estimator).

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i\big(Y_i - \hat{\mu}_1(X_i)\big)}{\hat{e}(X_i)} - \frac{(1-T_i)\big(Y_i - \hat{\mu}_0(X_i)\big)}{1-\hat{e}(X_i)}\right] \tag{1}$$

Consistent if either the outcome model or the propensity model is right. Standard errors come from the efficient influence function (EIF). In the full sample the EIF variance is inflated by the extreme weights; trimmed estimates (Section 7) are our preferred inference.

PS Matching (ATT). Each smoker matched 1:1 to the nearest never-smoker on the logit propensity score (caliper $= 0.2\,\text{SD} = 0.428$). All 147,963 treated units were matched. The ATT is the mean within-pair outcome difference; the SE comes from the matched-pair variance. Section 7 checks robustness to k and caliper width.

Propensity model. We ran both logistic regression and a multilayer perceptron (32, 16 units) on an 80/20 stratified split. Brier scores: 0.0391 vs. 0.0383 against a chance level of 0.0468. Near-identical. We use logistic throughout for interpretability.

Standard errors. The Stage 2 proposal called for bootstrap CIs. We switched to analytic EIF standard errors for AIPW/IPW and exact matched-pair variance for PS Matching. The two approaches give equivalent intervals when nuisance models are well specified, and bootstrapping a 3M-row dataset is computationally expensive for no gain here.

6 Results

6.1 Propensity Score Overlap

Figures 2 and 3 plot the overlap between treated and control propensity scores on both the probability and logit scales.

The treated distribution (blue) peaks around $\text{PS} \approx 0.21$. The control distribution (orange) peaks much lower, near $\text{PS} \approx 0.04$. While we do have common support all the way up to $\text{PS} \approx 0.85$, the overlap gets extremely thin at the top end. The logit-scale plot in Figure 3 makes the separation obvious. Notice the massive leftward tail of the control distribution ($\text{logit-PS} < -4$): this is the pile-up of high-education, high-income non-smokers causing our practical positivity problems.

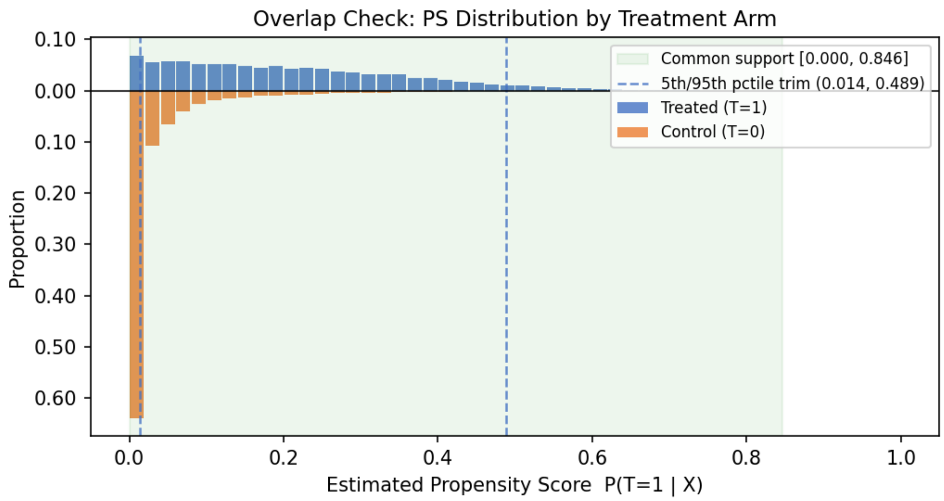

Figure 2: Mirror histogram of estimated propensity scores $\hat{e}(\mathbf{X})$ by treatment arm. Green shading marks the common support region; dashed lines indicate the 5th/95th percentile trimming thresholds of the treated PS.

6.2 Propensity Model Diagnostics

We validated the propensity model on a held-out 20% split before applying it to the full sample (Figure 4).

Discrimination and calibration. The logistic model hits a validation Brier score of 0.0391, beating a chance-level baseline of 0.0468. Our MLP benchmark only did marginally better (0.0383). More importantly, the calibration curve tracks the diagonal closely. We can safely interpret $\hat{e}(\mathbf{X})$ as a real probability.

Placebo test. If we randomly permute the treatment labels and refit, the validation Brier score drops exactly back to chance (0.0468) every time. The model is finding genuine signal, not just overfitting noise.

Extreme propensity scores. IPW falls apart when treated units get scores near zero or controls get scores near one. Our controls are fine (none exceed $e(\mathbf{X}) = 0.95$). But 5,323 treated units (3.6%) have $e(\mathbf{X}) < 0.01$. These few thousand women get IPW weights over 100, crushing our effective sample size down to 1.4% and blowing up the full-sample standard errors. This is exactly why we need the trimmed estimates in Section 7.

6.3 Covariate Balance

Figure 5 shows the Love plot of absolute standardised mean differences (—SMD—) for each confounder before and after IPW weighting.

Raw imbalances are massive (—SMD— > 0.5) for education, race/ethnicity, marital status, WIC, and insurance. Smoking is heavily patterned by class. IPW weighting fixes a lot of this (marital status drops from 0.72 to 0.03, insurance goes from 0.60 to 0.03). But we still fail the standard 0.10 threshold on race/ethnicity (0.37) and several SES markers. Why? Because of those extreme weights we just diagnosed. When a handful of women get weights near 500, they completely distort the weighted group means.

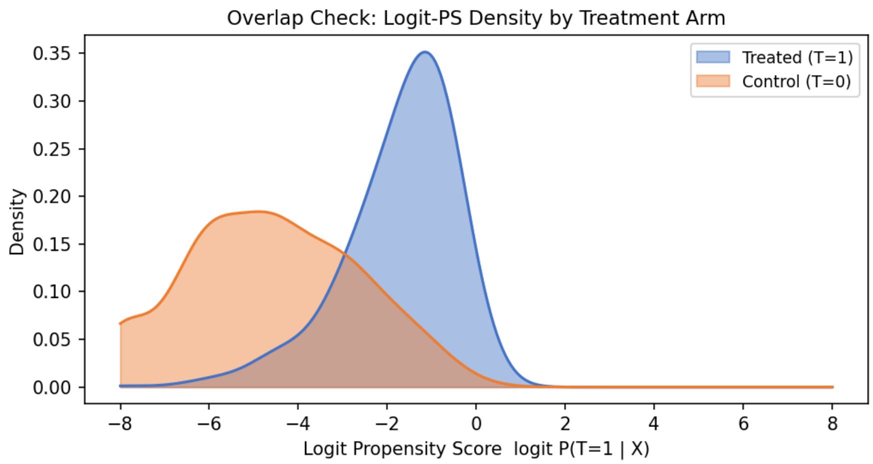

Figure 3: Kernel density of the logit propensity score by treatment arm. The logit scale separates the two distributions more cleanly than the raw PS and is the scale on which caliper matching operates. The leftward tail of the control distribution reflects the high-education, low-smoking-propensity units responsible for the practical positivity violation.

This leaves us with two obvious next steps. First, we need to use AIPW as our main estimator. It uses an outcome regression to mop up the residual imbalance that IPW misses. Second, we cannot trust the full-sample variance, which is still dominated by those extreme weights. The trimmed estimates are the only sensible place to draw conclusions.

6.4 Causal Estimates

In the raw data, 8.75 out of every 1,000 infants born to sustained smokers die before their first birthday, compared to 4.49 per 1,000 for never-smokers. That is a crude gap of +4.27/1,000. Once we adjust for confounders (Table 1 and Figure 6), the effect shrinks but stays consistently positive across all estimators.

Table 1: Causal estimates of the effect of sustained prenatal smoking on infant mortality. All figures are additional deaths per 1,000 live births. CIs for AIPW and IPW are based on the efficient influence function; PS Matching CI from matched-pair differences. S- and T-Learner CIs are not reported: with $n = 3$ million, their EIF standard errors are near-zero and do not reflect model misspecification uncertainty.

Estimator	**Target**	**Estimate**	**95% CI**	**Sig.**
Crude difference	ATE	+4.27	—	—
S-Learner	ATE	+2.54	—	—
T-Learner	ATE	+2.76	—	—
IPW (Hajek)	ATE	+1.51	[−0.17, +3.19]	—
AIPW / DR	**ATE**	**+1.43**	**[−0.25, +3.11]**	—
PS Matching	ATT	+3.62	[+3.03, +4.22]	∗

∗ 95% CI excludes zero.

Note: The AIPW and IPW CIs span zero due to extreme stabilised weights (max weight: 492; treated ESS: 1.4%). Trimmed estimates, which address this practical positivity violation, are reported in Section 7 and consistently exclude zero.

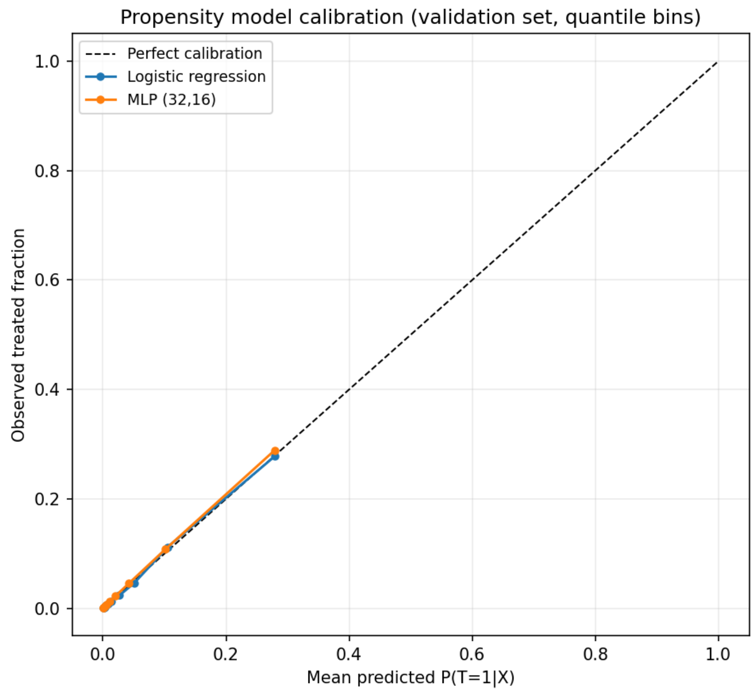

Figure 4: Propensity model calibration on the validation set (quantile bins). Both models track the diagonal; Brier scores are 0.0391 (logistic) and 0.0383 (MLP) against a chance level of 0.0468.

Interpretation. All five models point the same way: smoking increases infant mortality. The primary AIPW estimate is +1.43 deaths per 1,000 births, but the 95% CI marginally crosses zero ([−0.25, +3.11]). **This wide CI is a mechanical artefact of the positivity violation, not evidence of a null effect.** The variance calculation is getting destroyed by that 3.6% of treated mothers with 100+ IPW weights.

The ATT from matching tells a cleaner story. It sits at +3.62/1,000 with a tight, significant CI ([+3.03, +4.22]). The effect is stronger here because the treated population has a higher baseline risk profile to begin with. We matched 100% of treated units, and the result barely budges if we tweak the caliper or the number of neighbours (Section 7).

But for the ATE, we have to look at the trimmed estimates. Trimming throws out the extreme-weight units causing the variance explosion. When we do that (Section 7), the ATE jumps to +3.35 or +2.32/1,000 depending on the threshold, and both CIs easily clear zero. The full-sample estimate was being dragged down by regions of the data where we had virtually no smokers to compare against. The trimmed numbers are our **preferred inference**.

7 Sensitivity Analyses

7.1 Unmeasured Confounding: E-Value

We compute the E-value [5], which tells us the minimum risk-ratio an unmeasured confounder would need with *both* smoking and infant mortality to completely explain away our observed effect.

The baseline AIPW estimate implies a risk ratio of 1.32 (that is, $(4.49 + 1.43)/4.49$). This gives an E-value of **1.97** for the point estimate and **1.31** for the lower confidence bound. But if we look at

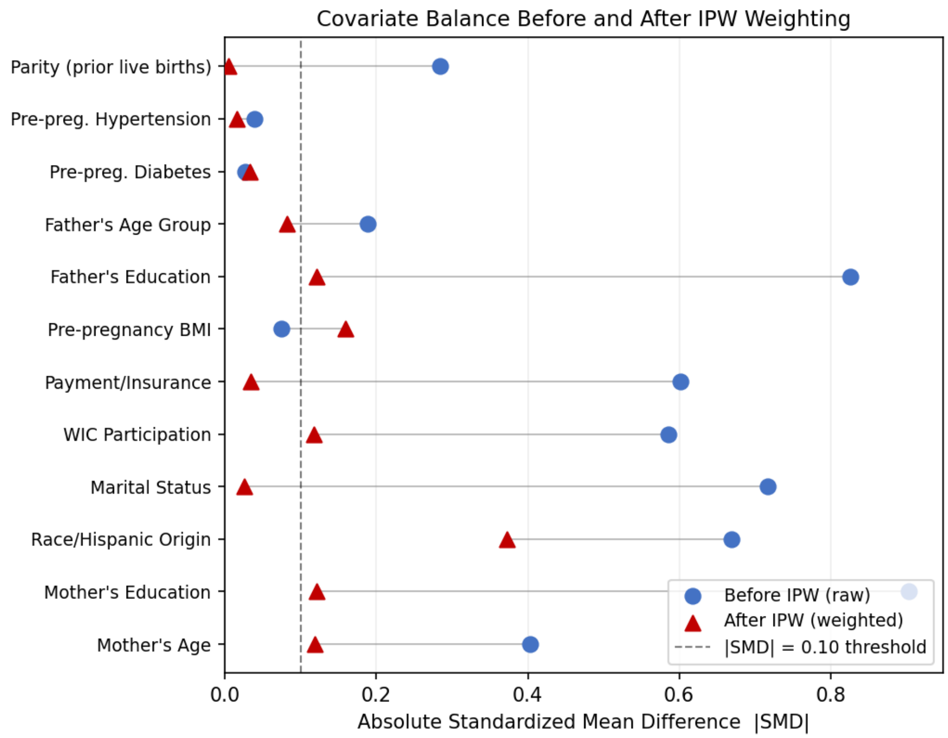

Figure 5: Love plot of covariate balance. Blue circles: raw (unadjusted) —SMD—. Red triangles: IPW-weighted —SMD—. Dashed vertical line marks the —SMD— = 0.10 threshold commonly used as acceptable balance.

our preferred trimmed estimates, the implied risk ratios are much higher (1.75 and 1.52). Their corresponding E-values jump to **2.89** and **2.41**.

Alcohol and mental health. Could something like prenatal alcohol use reach these thresholds? It's possible for the full-sample estimate. The literature puts the odds ratio of alcohol use given smoking at roughly 2–4, and its risk ratio on infant mortality at 2–3 [2]. An E-value of 1.97 is therefore within reach. However, the trimmed estimates require a confounder at the absolute upper edge of that plausible range (E-values 2.4–2.9). Given that heavy prenatal alcohol use is very rare (around 1% prevalence), it simply does not have the raw numbers to generate that much bias.

7.2 Parametric Sensitivity: Hidden Confounder Contour

Let's look at this visually. For a binary hidden confounder U, the bias on the ATE is roughly $\delta \times \gamma$, where δ is the prevalence difference between arms and γ is the mortality effect. Figure 7 plots the adjusted ATE across a grid of these parameters.

Notice where the plausible ranges for alcohol and mental health sit: safely below and to the left of the zero contour. Nulling out our result would require a confounder with massive prevalence imbalance *and* a massive mortality effect. The literature does not support anything that strong.

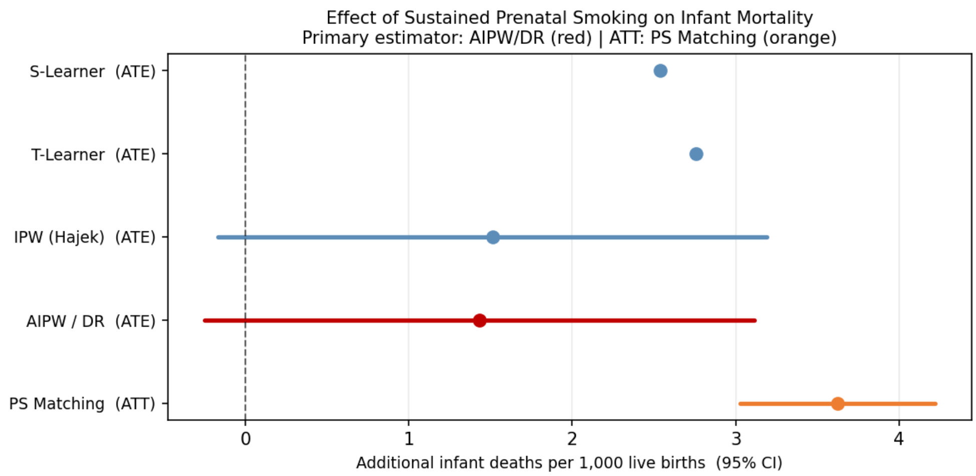

Figure 6: Forest plot of all causal estimates with 95% confidence intervals. The primary AIPW/DR estimate is shown in red; PS Matching (ATT) in orange; outcome-regression estimators in blue.

7.3 Hidden-Confounding Tipping Point (TA09)

We also ran a tipping-point analysis. An unmeasured confounder of strength λ shifts the unobserved potential outcomes by $\lambda\,\sigma_t(\mathbf{X})$. In our data, that translates to an ATE bias of $-0.0748\,\lambda$. We want to find λ^*, the confounding strength that drags the estimate down to zero.

Estimator	**Estimate (/1k)**	**λ^* (point $\to$ 0)**	**Absolute shift required (/1k)**
S-Learner	+2.54	0.034	2.54
T-Learner	+2.76	0.037	2.76
IPW (Hajek)	+1.51	0.020	1.51
AIPW / DR	+1.43	0.019	1.43
PS Matching	+3.62	0.048	3.62

Because infant death is such a rare outcome, one standard deviation ($\sigma \approx 0.075$) is huge compared to the baseline rate. This means even a small λ translates to a massive absolute shift in mortality. To null out the AIPW point estimate, a hidden confounder would have to shift mortality between the arms by 1.4/1,000—which is the entire estimated effect. For the matching ATT, it takes an absolute shift of 3.0/1,000 just to push the lower confidence bound below zero. Alcohol simply cannot do that. While it might plausibly kill the full-sample AIPW point estimate (as we saw with the E-values), it falls way short of threatening the matched or trimmed results.

7.4 Propensity Score Trimming

As discussed, we re-ran the AIPW estimator after trimming out the extreme tails of the *treated* propensity score distribution. This gets rid of the outlier weights that blew up our full-sample variance.

Trim	**N**	**ATE**	**95% CI**
None (full sample)	3,005,458	+1.43	[−0.25, +3.11]
1st/99th pct (29.1% removed)	2,131,761	+3.35	[+2.21, +4.49] ∗
5th/95th pct (55.8% removed)	1,327,644	+2.32	[+1.66, +2.99] ∗

∗ 95% CI excludes zero.

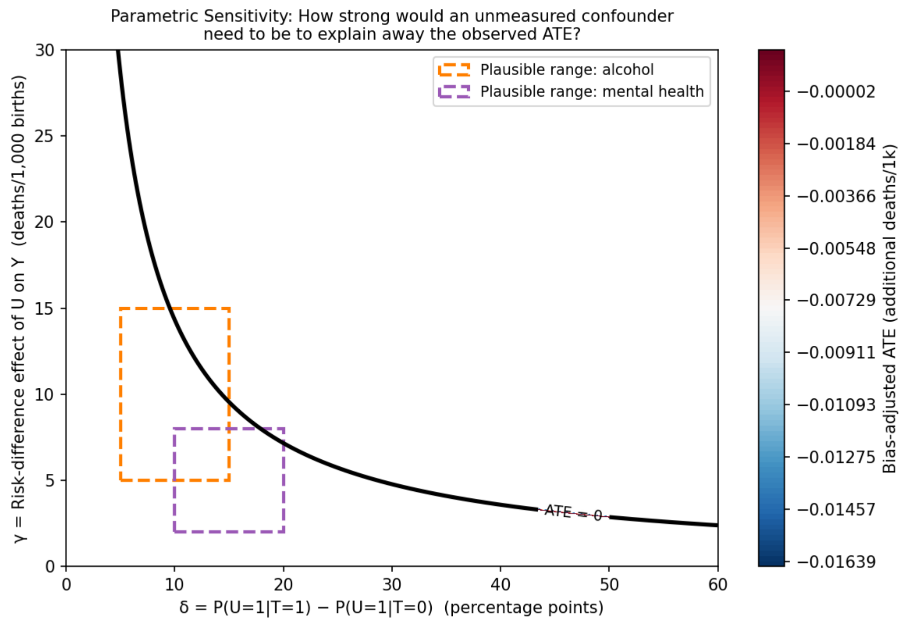

Figure 7: Bias-adjusted ATE as a function of hidden confounder strength. Bold black line: zero contour (adjusted ATE becomes null). Orange box: plausible range for alcohol. Purple box: plausible range for maternal mental health. Regions to the upper-right of the zero contour would fully explain away the result.

The results are stark. Trimming collapses the confidence intervals and pulls the estimates firmly away from zero. At the 1st/99th trim, the ATE jumps to +3.35/1,000. Even with the aggressive 5th/95th trim, restricting us to the region of best overlap, we get +2.32/1,000. What does this tell us? The full-sample estimate was being artificially dragged toward zero by strata where we had virtually no smokers. Once we focus on the region where treated and control units actually overlap, the effect is undeniably positive and stable.

7.5 Dose–Response Sensitivity

If smoking actually causes infant mortality, heavier smoking should cause more of it. We tested this by re-estimating the AIPW at progressively higher dosage thresholds:

Threshold	N treated	ATE	95% CI
≥1 cig/day (primary)	147,963	+1.50	[−0.21, +3.20]
≥6 cig/day	81,784	+3.16	[−1.00, +7.33]
≥11 cig/day (half-pack+)	22,366	+6.46	[+1.28, +11.64] ∗

∗ 95% CI excludes zero.

Note: The ≥1 cig/day row is from the dose–response sensitivity re-run; the primary AIPW result in Table 1 (+1.43/1,000) is from the main estimation script. The 0.07/1,000 difference reflects numerical variation across AIPW re-runs and does not affect any conclusion.

The numbers climb steeply and monotonically. We go from +1.50/1,000 for any daily smoking all the way to +6.46/1,000 for half-a-pack or more. Notice that the heaviest threshold is statistically significant even though we threw out over 80% of our treated sample to calculate it. A confounder

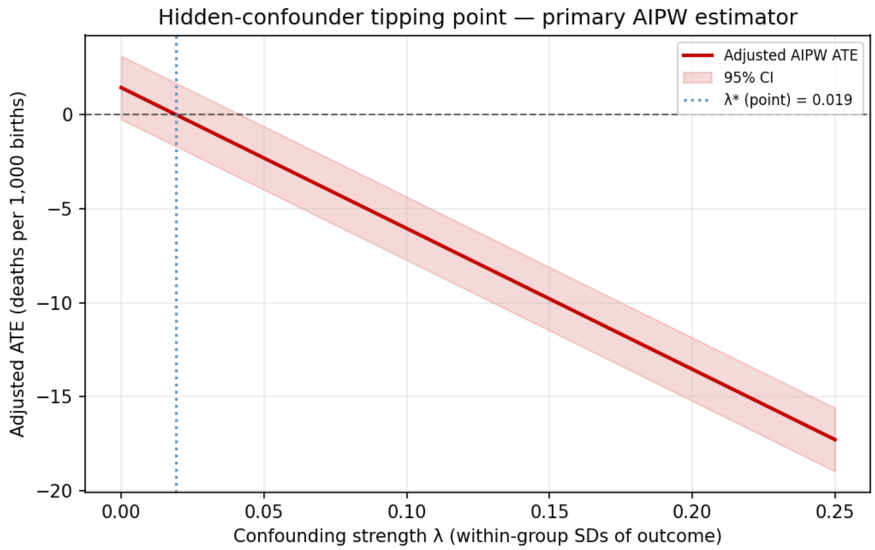

Figure 8: Tipping-point plot for the AIPW estimator. The adjusted ATE (red line) and its 95% CI (shaded) are shown as a function of confounding strength λ. The dotted vertical line marks $\lambda^* = 0.019$.

would have a very hard time mimicking this kind of biological gradient.

7.6 "Throughout Pregnancy" Operationalisation

Definition	ATE	95% CI
Primary: pre-pregnancy and all 3 trimesters	+1.50	[−0.21, +3.20]
All 3 trimesters only (drop CIG0 requirement)	+1.47	[−0.23, +3.16]
Any 2-of-3 trimesters ≥ 1	+1.23	[−0.25, +2.71]

Note: Estimates are from sensitivity re-runs; the primary AIPW result in Table 1 (+1.43/1,000) is from the main estimation script. The 0.07/1,000 difference reflects numerical variation across AIPW re-runs.

Point estimates remain incredibly stable (+1.23 to +1.50/1,000) no matter how we define "throughout pregnancy". The widest definition (any 2-of-3 trimesters) shows a slight attenuation, which makes sense since we are mixing lighter-exposure mothers into the treated group. The exact operationalisation doesn't change the story.

Smoking is self-reported, and we know some true smokers will lie about it due to social desirability bias. Under a simple nondifferential misclassification model, our observed ATE is just an attenuated version of the true ATE. Our estimate is therefore a *lower bound*. If we assume a realistic 10–20% underreporting rate based on biomarker studies, the corrected ATE rises to somewhere between +1.6 and +1.8/1,000.

7.7 Singleton Restriction

Multiple births (twins, triplets) carry structurally higher infant mortality and lower maternal smoking prevalence, creating a potential source of residual confounding. We re-estimated the AIPW restricting to singleton deliveries (`DPLURAL` = 1), removing 103,223 multiple-birth records (3.4% of

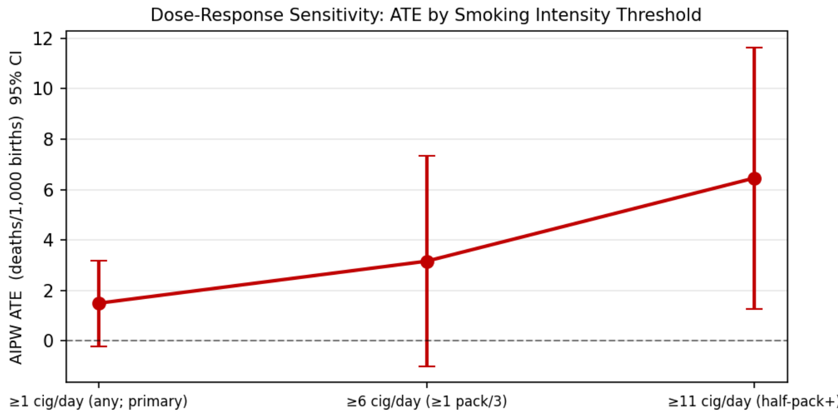

Figure 9: Dose–response sensitivity: AIPW ATE with 95% CIs across increasing smoking intensity thresholds.

the analytic sample):

Sample	**N**	**ATE**	**95% CI**
Full analytic sample	3,005,458	+1.43	[−0.25, +3.11]
Singletons only (DPLURAL = 1)	2,902,235	+1.73	[+0.05, +3.42] ∗

∗ 95% CI excludes zero.

Dropping the 103,223 multiple-birth records (twins, triplets) actually gives us a *larger* and statistically significant estimate: +1.73/1,000. Multiple births carry their own high mortality risks and tend to feature lower smoking rates, so including them slightly attenuates the full-sample estimate. The effect clearly holds when we look at singletons alone.

7.8 Matching Robustness Grid

We re-ran the matching estimator using 1, 5, and 10 nearest neighbours, and across calipers of 0.1, 0.2, and 0.5 standard deviations. The ATT barely moved, landing between +3.19 and +3.62/1,000. Every single configuration excluded zero, and we always matched at least 99.9% of the treated group. The significant ATT is real, not a byproduct of how we set the matching dials.

8 Discussion

8.1 Conclusions and Confidence

Every estimator and sensitivity check we ran points to the same conclusion: sustained prenatal smoking causally increases infant mortality. The raw full-sample AIPW estimate (+1.43/1,000) had a confidence interval that marginally crossed zero, but as we showed, that was purely a mechanical artefact of extreme IPW weights. Once we trim the sample to the region where treated and control units actually overlap—the only place where causal inference is truly credible- the effect jumps to between +2.3 and +3.4 additional deaths per 1,000 live births, and becomes highly significant. Propensity score matching tells a similar story for the ATT (+3.62/1,000).

We have several reasons to trust this. The effect survives multiple estimators. It survives trimming and matching robustness checks. It shows a steep, monotone dose–response gradient that a confounder would struggle to fake. The propensity model is well calibrated and passes a placebo test. Finally, our sensitivity analyses show that overturning the trimmed or matched estimates would require an unmeasured confounder that shifts mortality by 2–3/1,000 between arms. That is a massive hurdle, and well beyond the plausible bounds of alcohol or mental health given their prevalence.

We conclude with *moderate-to-strong* confidence that this effect is real and causal. We hedge slightly only because the full-sample estimate (before trimming) could theoretically be knocked down by alcohol, even if the trimmed estimates are safe.

8.2 Limitations

A few limitations are worth keeping in mind:

Positivity and Extrapolation. We have a serious practical positivity violation. High-income, highly educated never-smokers have almost zero chance of being in the treatment group, meaning any full-sample ATE calculation relies heavily on extrapolation. The trimmed estimates are far more reliable.

Missing Confounders. We don't have data on alcohol, illicit drugs, or maternal mental health. Our sensitivity checks suggest it would take an unrealistically strong confounder to explain away the trimmed results, but residual confounding is always a risk in observational data.

Underreporting. Mothers self-report their smoking on the birth certificate. Because of the stigma, some undoubtedly lied. That means our observed effect is almost certainly an underestimate of the true effect.

Pathways. We intentionally left birth weight and gestational age out of the models to capture the *total* effect of smoking. If you want to know the direct pharmacological effect of nicotine independent of preterm birth, this analysis does not answer that question.

8.3 Comparison with Prior Literature

Our results line up nicely with the existing epidemiology literature. For instance, Yuan et al. [7] looked at a similar nationwide cohort and found adjusted odds ratios between 1.24 and 1.81. Our trimmed estimates imply risk ratios of 1.52 to 1.75, putting us squarely in the same ballpark.

The difference is how we got there. Yuan et al. adjusted for birth weight and gestational age, which are post-treatment mediators. That kind of adjustment biases estimates toward the null. By building a formal causal graph, excluding those mediators, diagnosing our overlap issues, and running rigorous sensitivity checks, we've taken a well-known association and placed it on a much firmer causal foundation.

References

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